Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/31/2025
PI Name:	Kim, Han Ph.D.		
Project Title:	3D Scanning System to Assess Gravity-Dependent Body Shape Changes		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (2) **Dynamic Loads**:Risk of In-Mission Injury and Performance Decrements and Long-term Health Effects due to Dynamic Loads (3) **EVA**:Risk of Mission Impacting Injury and Compromised Performance and Long-Term Health Effects due to EVA Operations (4) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77002	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2023	**End Date:**	09/30/2024
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Sudhakar Rajulu retired in 2024, thus were removed from the Co-Investigator list. Nate Newby was added to the Co-Investigator list as JSC-SK/AIBEL Technical Discipline Lead Linh Vu and Karen Young were removed the project, due to the HHPC task order adjustments Pouyan Sabahi and Dan Nguyen were added as Co-Investigator due to the HHPC task order adjustments		
COI Name (Institution):	Sabahi, Pouyan (KBR, Inc.) Nguyen, Dan (Leidos, Inc.) Newby, Nate (KBR, Inc.)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The human body shows unique physiological and morphological changes in different gravity conditions, including muscle loss, fluid shift, spinal elongation, and body posture adjustments. This work aims at developing a 3D scanning system for crewmembers' body shapes, of which the design and operations are optimized for reduced gravity aircraft, space habitats, or vehicles. A hardware system will be custom-built using a network of commercial infrared sensors. Data acquisition and processing software will be developed for efficient comparisons across pre-, in- and post-flight phases. The system will be tested in the ground laboratory for accuracy and reliability against the reference scanning system.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. 3D body scanning technology has not yet been made available for spaceflights or reduced gravity analogues.Another constraint is system calibration. To overcome the limitations of the existing scanners, the JSC Anthropometry & Biomechanics Facility (ABF) has inquired commercial vendors about the possible modification of commercial scanners but was informed that scan systems need to be redesigned entirely, which can be costly, to meet the requirements for spaceflight use.
Research Impact/Earth Benefits:	This work proved the feasibility of 3D body scanner development, custom designed and optimized for space vehicle and habitat deployment and operation. The system was built using a set of commercial off the shelf (COTS) sensors, reducing the development and implementation costs, potentially for other ground-based applications, such as medical imaging or garment fit. A new technique was identified and developed to calibrate the body scanning system, which overall can substantially reduce time and effort overhead. With an increased technology readiness level, the system has the potential to mature and be deployed to capture the anthropometric changes in spaceflight, of which the data can provide critical information to improve human-system integration and mitigate of some crew health issues, such as low back pain. The knowledge and data acquired from this work can inform future development and operation of in-flight body scanning.
Task Progress:	The human body shows unique morphological changes when exposed to different gravity conditions, including muscle atrophy, fluid shift, and spinal elongation. Such changes need to be incorporated for human-system integration in the vehicle habitat, garment, and spacesuit designs, as inaccurate body measurements can result in suboptimal crew protection that can potentially decrease injury tolerance. However, measurement tools have not been available for accurate assessments of body shape changes. This work aimed to develop a prototype 3-D body scanning system with the configuration and performance optimized for inflight crewmember body scanning. A scan hardware system was developed using Intel RealSense commercial off-the-shelf 3D sensors. The sensor parameters were iteratively optimized and tested to obtain the performance level needed for body scanning. A scan booth structure was fabricated, with the overall size 4 x 4 x 8 feet. The specific number of sensors and mounting positions were determined by iterative simulations, which indicated that 16 cameras can capture 94% and 96% of body surface area from the 1st percentile female and 99th percentile male crew population subject. The mounted sensors were linked through a mix of USB-C and USB-3 cables and operated for data acquisition from a Linux laptop computer. A software prototype was developed using Python and Tkinter graphical user interface toolkit. A calibration procedure was also built using a panel of QR codes. A computer vision tool detected and decoded the unique ID and pattern locations of the QR codes. The calibration information determined the position and orientation of the 3D sensors with respect to each other. The scanner performance was assessed using a set of 3D printed custom manikins. The manikin size and shape were derived from the previous International Space Station (ISS) study, which measured the crewmembers' anthropometry changes across the different flight phases. The average anthropometric measurements at the preflight and flight day 15 were sampled and projected onto the 1st percentile female and 99th percentile male body shapes. Another pair of manikins was also 3D printed to simulate the neutral body posture, estimated from ISS microgravity environments. A preliminary analysis assessed the performance of the newly developed scanner against the reference scanner, which has been used at the NASA Johsnson Space Center (JSC) for crew and test subject anthropometry. Although the new scanner data showed several artifacts and missing geometries in the occluded body areas such as armpits and crotch, overall shape matched with the reference scan. When the manikin surface coordinates were compared, a root mean square error of 1.3 cm was observed from the manikin torso segment. Linear measurements including the stature, knee height, and circumference measurements at the chest and calf showed differences from the reference scan measurements, ranging between 0.3 and 0.9 cm. Overall, this work demonstrated a development framework for an inflight scanner with design and operation optimized for crewmember body scanning. Further improvement can potentially provide previously unavailable anthropometric data from different gravitational environments, including 0-g, 1/6-g, and 1-g. Such data can improve suit fit, habitat design, exercise efficacy quantification, and sizing of orthostatic intolerance garments.
Bibliography Type:	Description: (Last Updated: 06/22/2026)
Abstracts for Journals and Proceedings	Kim KH, Sabahi SP, Nguyen D, Newby N. "3D scanning system to assess gravity-dependent body shape changes." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Poster. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Kim KH, Sabahi SP, Nguyen D, Vu LQ, Newby N. "3D scanning system to assess gravity-dependent body shape changes." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Poster. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Articles in Peer-reviewed Journals	Green W, Kim H, Benson E, Rajulu S. "Spacesuit contact with wearer: Analysis and visualization." Acta Astronaut. 2024 Sep 1;222:450-6. https://doi.org/10.1016/j.actaastro.2024.06.003 , Sep-2024